User Study – Manual Python Workflow (Method 1 Only)

Instructions: Fill Section A once. For Section B, answer *per use-case*. Duplicate Section B if you completed multiple use-cases.

Participant ID: use case2-user1 3 **Your assignment:** ☑ **Method 1** (Manual Python)
Use case(s) you worked on: ☐ U1 River Flow ☑ U2 Smart-Home Energy ☐ U3 Solar Energy

Section A – Background (answer once)

A1. Educational level *(highest completed)*
☐ High School ☐ Vocational/Technical ☐ Bachelor's ☑ Master's ☐ Doctorate ☐ Other: ____________

A2. Primary role *(pick the closest)*
☐ IoT/CPS Developer ☐ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☑ Software Architect / Full-stack ☐ System Integrator ☐ Other: ____________

A3. Self-rated expertise *(tick one per row)*

	Beginner	Intermediate	Advanced	Expert
Python programming	☐	☐	☐	☑
Time-series / ML	☐	☐	☑	☐
Domain knowledge (your use-case)	☐	☑	☐	☐

A4. Prior experience *(used ML frameworks/libraries before?)* ☑ Yes ☐ No

Section B – Task-Specific (duplicate this page *per use-case*)

This B-section is for: ☐ U1 River Flow ☑ U2 Smart-Home Energy ☐ U3 Solar Energy

B1. Time & Effort

Record your own times. Estimate if needed. "Extra learning time" = time spent reading docs/tutorials.

B1. Time spent (minutes)

	Method 1 (Manual Python)
Writing/testing code (end-to-end)	200 minutes
Extra learning time	100 minutes

B2. Lines of code (LoC) *(exclude blanks/comments)*
Manual Python LoC: ~120

B2. Task Completion & Ease of Use

Completion: None = could not finish; Partial = finished some parts; Complete = all parts.

B3. Were you able to finish the use-case with Manual Python?
Method 1 (Manual Python): ☐ None ☐ Partial ☑ Complete

B4. Ease by workflow step (Manual Python) *(1=Very easy, 5=Very difficult; tick one per cell)*

Step	1	2	3	4	5
Whole workflow	☐	☐	☑	☐	☐
Data preprocessing	☐	☑	☐	☐	☐
Feature engineering (lags/rolling)	☑	☐	☐	☐	☐
Model training	☐	☐	☐	☑	☐
Model evaluation/plots	☐	☐	☑	☐	☐

B1. Which steps did you complete? *(check all that apply)*

☑ **Preprocessing**
☑ **Model configuration**
☑ **Training**
☑ **Evaluation**
☑ **Plotting/Diagnostics**

B2. How difficult was each step? *(1=Very easy, 5=Very difficult; tick one per row)*

Step	1	2	3	4	5
Preprocessing	☐	☑	☐	☐	☐
Model configuration	☐	☐	☐	☑	☐
Training	☐	☐	☐	☑	☐
Evaluation	☐	☐	☑	☐	☐
Plotting/Diagnostics	☐	☐	☑	☐	☐

B3. Notes (issues, decisions, or gaps to revisit):

Resampled 1-minute data to 15-minute intervals and simplified SARIMA/SARIMAX to (1,1,0) × (1,1,0,96) to reduce execution time; skipped daily repeat baseline due to initial data size assumption. Revisit 1-minute cadence, daily repeat, and complex model orders with full dataset for better accuracy.

B5. Time spent per step *(record for this use-case; minutes or start/end)*

Step	**Start (hh:mm)**	**End (hh:mm)**	**Minutes**
Preprocessing	______	______	20 minutes
Model configuration	______	______	30 minutes
Training	______	______	40 minutes
Evaluation (metrics/baseline)	______	______	20 minutes
Plotting / Diagnostics	______	______	20 minutes
Debugging (errors, crashes, outputs)	______	______	30 minutes
Extra learning / searching / docs	______	______	90 minutes
Total			250 minutes

Notes: "Learning" includes reading docs, web searches, tutorials, setup notes, and guides. "Debugging" covers error fixing, crashes, failed runs, and investigating unexpected results.

B6. Comments (bugs, friction points, ideas):

Bugs: Encountered NameError: rmse_sarima due to undefined variable and ValueError: x and y must have same first dimension from mismatched prediction dimensions, resolved through debugging.
Friction Points: Long execution times with 1-minute data and seasonal period 1440; resampled to 15-minute for feasibility. Daily repeat baseline omitted due to initial data size assumption.
Ideas: Use parallel processing (e.g., joblib) to speed up SARIMA/SARIMAX. Test 1-minute cadence with full dataset and add daily repeat baseline for complete comparison.

Train Holt–Winters 5min Test 2min

Train SARIMA 10min Test 3min
Train /SARIMAX 15min Test 5min